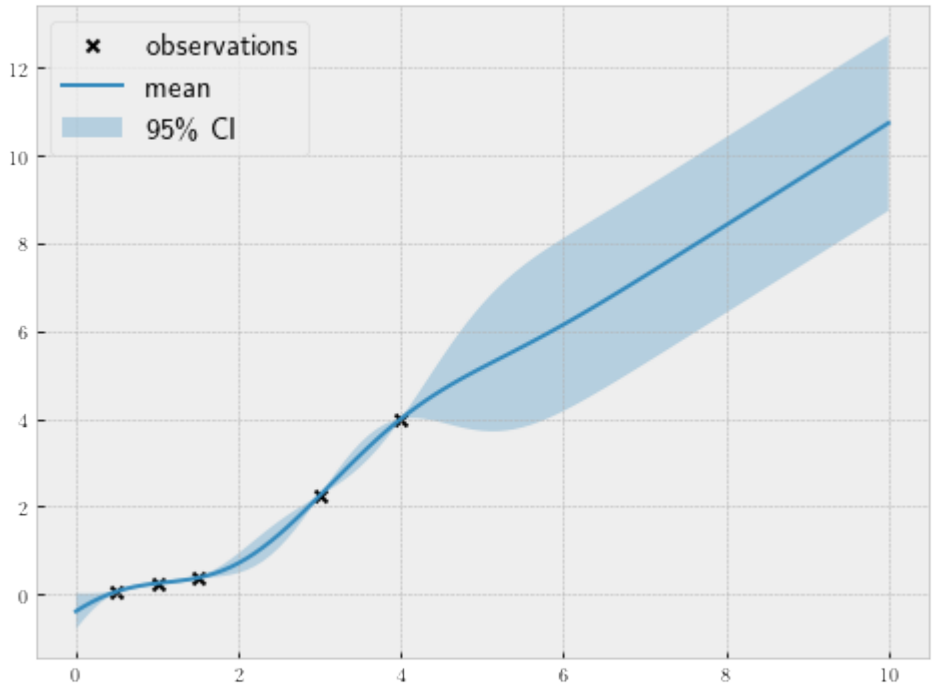


Figure 3.8 Predictions by a GP with a linear mean function. The GP has an upward trend, which is a direct result of the positive slope of the linear mean function.

Unlike what we have seen so far with the constant mean, the linear mean function we're using here drives the entire GP model to have an upward trend. This is because the best-fit line of the five data points in our training data is one with a positive slope, which is exactly what we wanted to model as a relationship between living area and price.

3.3.4 Using the quadratic function by implementing a custom mean function

Our linear mean function here successfully captures the increasing trend of the price, but it assumes that the rate of price increase is constant. That is, adding an extra squared foot to the living area leads to, in expectation, a constant increase in price.

In many cases, however, we might have access to the prior knowledge that our objective function increases at a non-constant rate, which a linear mean cannot model. In fact, the data points we've been using were generated so that the price is a quadratic function of the living area. This is why we see the larger houses become more expensive faster than the smaller houses. In this subsection, we will implement our GP mean as a quadratic function.

At the time of this writing, GPyTorch only supports constant and linear mean functions. But the beauty of this package, as we will see again and again in the book, is its modularity: all components of a GP model such as the mean function, the covariance function, the prediction strategy, or even the marginal log likelihood function, are implemented as modules, and therefore can be modified, re-implemented, and extended in an object-oriented manner. We will see this first-hand when implementing our own quadratic mean function.

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